EE1007
Introduction to Computer Science Laboratory

Lab 11
Pointer

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Lab 1.1
P113037. Price Discount(30%)

- Test target: pointer
- You are asked to write a program to calculate the discounted price. First, enter the original price and the discount. Then, calculate the price to the second decimal.

Lab 1.1
P113037. Price Discount(30%)

◆思考方向



Lab 1.1
P113037. Price Discount(30%)

◆ Assign the address to the pointer

```
std::cout << "Please enter the discount: ";
std::cin >> discount;
// DON'T REMOVE THIS LINE
// YOU CAN ONLY WRITE HERE!!!
// CODE BEGIN
pricePtr = &price;
// CODE END
// DON'T REMOVE THIS LINE
```

◆ Calculate the price

```
// DON'T REMOVE THIS LINE
// YOU CAN ONLY WRITE HERE!!!
// CODE BEGIN
price = price * discount;
pricePtr = &price;
// CODE END
// DON'T REMOVE THIS LINE

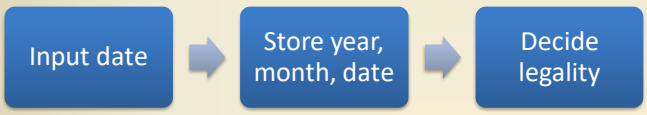
std::cout << "After discount, the price is " << *pricePtr << std::endl;
```

Lab 11.2
P113038. Date(30%)

- Goal: array
- You are asked to write a program to decide whether the date is legal or not.
- First, the user will enter the date, ex:20240807. For the example date, year=2024, month=8, day=7. Then the program will print “This is legal.”, otherwise print “This is not legal.”.

Lab 11.2
P113038. Date(30%)

- 思考方向



Lab 11.2
P113038. Date(30%)

- Change char to int

```
for (int i=0;i<8;i++) {
    cin >> date[i];
}

for (int i=0;i<8;i++) {
    date_int[i] = int(date[i]) - 48;
}
```

- Store year, month and day

```
day = date[6]*10+date[7];
month = date[4]*10+date[5];
year = date[0]*1000+date[1]*100+date[2]*10+date[3];
```


Lab 11.2
P113038. Date(30%)

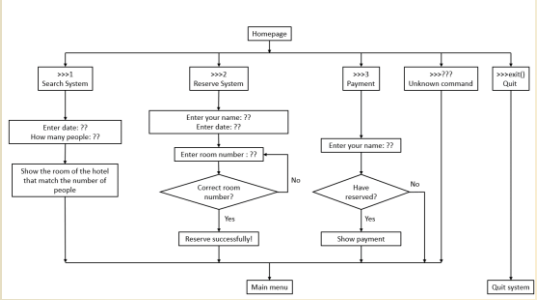
- Decide leap year

```
case 2:
    if(day<1) result = false;

    if(year%4!=0){//not leap year
        if(day>28)result = false;
        else result = true;
    }
    else if(year%4==0 && year%100!=0 ){//leap year
        if(day>29)result = false;
        else result = true;
    }
    else if(year%400==0 ){//leap year
        if(day>29)result = false;
        else result = true;
    }
    else if(year%1000==0 ){//leap year
        if(day>28)result = false;
        else result = true;
    }
    else{//not leap year
        if(day>28)result = false;
        else result = true;
    }
    break;
```

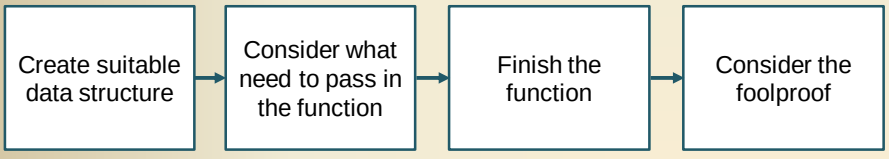
Lab 11.3
P113039. Booking System(30%)

- Test target: function
- You are asked to write a program for a booking system. There are three main functions in this program.



Lab 11.3
P113039. Booking System(30%)

- 思考方向



Lab 11.3
P113039. Booking System(30%)

◆ Search System

```
if (people == 2 || people == 4)
    cout << "*** " << left << setw(20) << "Cloud Hotel" << left << setw(19) << "Distance: 0.6km ***" << endl;

if (people == 2 || people == 4)
    cout << left << setw(4) << "No." << left << setw(35) << "Room" << left << setw(5) << "Price" << endl;

if (people == 2)
    cout << left << setw(4) << "1-1" << left << setw(35) << "Double" << "NTD " << right << setw(
            4) << "1650" << endl;

if (people == 4)
    cout << left << setw(4) << "1-2" << left << setw(35) << "Quad" << "NTD " << right << setw(
            4) << "3200" << endl;

if (people == 2 || people == 4)
    cout << endl;
```

```
if (people == 1 || people == 2 )
    cout << "*** " << left << setw(20) << "Garden Hotel" << left << setw(17) << "Distance: 1km ***" << endl;

if (people == 1 || people == 2)
    cout << left << setw(4) << "No." << left << setw(35) << "Room" << left << setw(5) << "Price" << endl;

if (people == 1)
    cout << left << setw(4) << "2-1" << left << setw(35) << "Single" << "NTD " << right << setw(
            4) << "760" << endl;

if (people == 2)
    cout << left << setw(4) << "2-2" << left << setw(35) << "Double" << "NTD " << right << setw(
            4) << "1380" << endl;

if (people == 2)
    cout << left << setw(4) << "2-3" << left << setw(35) << "Twin" << "NTD " << right << setw(
            4) << "1420" << endl;

if (people == 1 || people == 2 )
    cout << endl;
```


Lab 11.3
P113039. Booking System(30%)

◆ Search System

```
if (people >= 2)
    cout << "*** " << left << setw(20) << "City Big Hotel" <<  left << setw(19) << "Distance: 1.5km ***" << endl;

if (people >= 2)
    cout << left << setw(4) << "No." << left << setw(35) << "Room" << left << setw(5) << "Price" << endl;

if (people == 2)
    cout << left << setw(4) << "3-1" << left << setw(35) << "Double" << "NTD " << right << setw(
            4) << "2540" << endl;

if (people == 3)
    cout << left << setw(4) << "3-2" << left << setw(35) << "Triple" << "NTD " << right << setw(
            4) << "3670" << endl;

if (people == 4)
    cout << left << setw(4) << "3-3" << left << setw(35) << "Quad" << "NTD " << right << setw(
            4) << "4300" << endl;

if (people >= 2)
    cout << endl;
```

Lab 11.3
P113039. Booking System(30%)

- Reserve System

```
for (int i = 0; i < number_people; i++) {
    if (reserve[i][0] == name) {
        for(int j=0;j<50;j++){
            if(reserve[i][j]==""){
                hold = j;
                break;
            }
        }
        Rooms(temp, input);
        reserve[i][hold] = date;
        reserve[i][hold + 1] = temp[0]; //room
        reserve[i][hold + 2] = temp[1]; //price
        reserve[i][hold + 3] = input;
        cout << endl;
        return number_people;
    }
}

//name no reserved before
reserve[number_people][0] = name;
reserve[number_people][1] = date;
Rooms(temp, input);
reserve[number_people][2] = temp[0]; //room
reserve[number_people][3] = temp[1]; //price
reserve[number_people][4] = input;
number_people++;
```

Get room information

Lab 11.3
P113039. Booking System(30%)

- Payment

```
for (int i = 0; i < number_people; i++) {
    if (reserve[i][0] == name) {
        cout << endl;
        cout << "*** This is " << name << "'s order ***" << endl;

        for (int j = 1; j < 21;j++) {
            if(reserve[i][j]!=""){
                cout << "Date: " << reserve[i][j] << endl;
                cout << left << setw(35) << "Hotel" << left << setw(10) << "Price" << endl;
                cout << "----------------------------------------" << endl;

                cout << left << setw(35) << reserve[i][j + 1] << "NTD " << right << setw(4) << reserve[i][j + 2] << endl;
                sum += stoi(reserve[i][j + 2]);
                j += 3;
                cout << endl;
            }

        }

        cout << "----------------------------------------" << endl;
        cout << left << setw(35) << "Total" << "NTD " << right << setw(4) << sum << endl;
        cout << endl;
        return;
    }
}
```

Hotel room's name

price

Lab 11.3
P113039. Booking System(30%)

- How to call 2D array

```
string reserve[50][21];
int number_people = 0;
```

```
else if (order == "2") {
  cout << endl;
  number_people = Reserve_System(reserve,number_people);
```

```
int Reserve_System(string reserve[][21], int number_people) {
```


Lab 11.3
P113039. Booking System(30%)

- Price and NTD need to be aligned

```
*** Cloud Hotel          Distance: 0.6km ***
No. Room                                  Price
1-1 Double                                NTD 1650

*** Garden Hotel         Distance: 1km   ***
No. Room                                  Price
2-2 Double                                NTD 1380
2-3 Twin                                  NTD 1420

*** City Big Hotel       Distance: 1.5km ***
No. Room                                  Price
3-1 Double                                NTD 2540
```

```
Date: 9/6
Hotel                                     Price
-----------------------------------------------
City Big Hotel Double Room                NTD 2540

Date: 2/10
Hotel                                     Price
-----------------------------------------------
Garden Hotel Twin Room                    NTD 1420

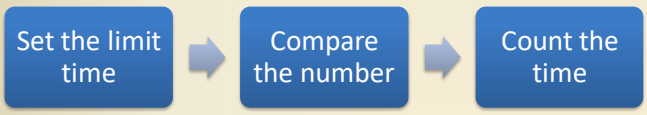
-----------------------------------------------
Total                                     NTD 3960
```

Lab 11.4
P113040. Apple and Banana (20%)

- Goal: array
- You are asked to realize a game to guessing secret number. First, the program will ask the user to enter the lenth of the secret number. All the digit of the number will not repeat. Next, the program will ask the user to enter the secret number. Then, the player starts to guess. The program will give the hint to the player by the following rules:
  - Apple: The number of digits that are correct in both digit and position.
  - Banna: The number of digits that are correct in digit but not in the correct position.
- Finally, if the player got length Apple and 0 Banna within limited times, the player wins; otherwise, the player loses.

Lab 11.4
P113040. Apple and Banana (20%)

- 思考方向



Lab 11.4
P113040. Apple and Banana (20%)

- Store limited time

```
int lim_time[6][2] = {{3,4}, {4,7}, {5,11}, {6,16}, {7,22}, {8,29}};
```

- Count A → same position and number

```
for(int i=0;i<length;i++){
    if(g[i]==n[i]) counterA += 1;
    else{
        nkeep[B_num] = n[i];
        gkeep[B_num] = g[i];
        B_num++;
    }
}
```

- Count B → same number but different position

```
for(int i=0;i<B_num;i++){
    for(int j=0;j<B_num;j++){
        if(nkeep[i]==gkeep[j]) counterB++;
    }
}
```


Evaluation

- Lab11.1(30%)
  - 5 testcases, 6% each
- Lab11.2(30%)
  - 5 testcases, 6% each
- Lab11.3(30%)
  - 5 testcases, 6% each
- Lab11.4(20%)
  - 5 testcases, 4% each





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